

Wangbiao Guo, Ph.D.

Yale West Campus | 840 West Campus Dr, ABC 211 | West Haven, CT 06516-7389
203-390-0472 | wangbiao.guo@yale.edu

Research Interests

In situ cryo-EM | Cryo-ET | Cryo-FIB | FIB-SEM | Cellular nanomachines | Bacterial motility
Bacterial signaling | Host-pathogen interactions | Biofilm | Eukaryotic cells | Tissues

Education

2021 **Ph.D., Energy and Environmental Engineering**, State Key Laboratory of Clean Energy Utilization and Center of Cryo-Electron Microscopy, Zhejiang University, China
2016 **B.S., Safety Engineering**, China University of Mining and Technology, China

Research Experience

09/2021–
Present **Postdoctoral Associate, Lab of Prof. Jun Liu**
Department of Microbial Pathogenesis, Yale School of Medicine, New Haven, CT
Research focus: *In situ* single-particle cryo-electron microscopy (cryo-EM SPA), cryo-electron tomography (cryo-ET), and cryo-focused ion beam (cryo-FIB) milling of living microbes.
Projects: *In situ* near-atomic structures of flagella in *Vibrio cholerae* and *Borrelia burgdorferi*.
Aquilos2 Manager
Responsibilities: Develop protocols, consult with engineers, perform equipment maintenance, train researchers, and troubleshoot issues.
Projects: Biological specimens and bacteria, including HeLa cells, sperm, red blood cells, *Chlorella*, *Nannochloropsis salina*, *Proteus mirabilis*, *Bacillus subtilis*, *Vibrio cholerae* biofilms, and cyanobacteria.

09/2019–
03/2020 **Postgraduate Fellow, Lab of Prof. Jun Liu**
Department of Microbial Pathogenesis, Yale School of Medicine, New Haven, CT
Research focus: Cryo-electron tomography and cryo-focused ion beam milling of microalgal cells, including *Chlorella* and *Nannochloropsis salina*.

09/2018–
06/2021 **Ph.D. Student, Lab of Prof. Xing Zhang**
Center of Cryo-Electron Microscopy, Zhejiang University, China
Research focus: Focused ion beam scanning electron microscopy (FIB-SEM) of high-pressure-frozen *Chlorella pyrenoidosa* cells.

09/2016–
06/2021 **Ph.D. Student, Lab of Prof. Jun Cheng**
State Key Laboratory of Clean Energy Utilization, Zhejiang University, China
Research focus: Bioengineering technology development for flue-gas CO₂ fixation by microalgal cells, including nuclear-radiation mutagenesis, photobioreactor design, and large-scale engineering processes.

Selected Publications

1. **Guo, W.^{*#}**, Zhang, S.^{*}, Park, J., Stanton, V., Asp, M., Herrera, H., Tai, B., Yue, J., Wang, J., Guo, J., Kumar, R., Botting, J., Wu, S., Yan, J., Klose, K., Yildiz, F., & Liu, J.[#] (2025). Structures of the sheathed flagellum reveal mechanisms of assembly and rotation in *Vibrio cholerae*. **Nat. Microbiol.**, 10, 3305–3314.
News: *New View of Cholera ‘Tail’ Could Inform Treatment.* Yale School of Medicine.

2. **Guo, W.**^{*}, Yue, J.^{*}, Park, J.^{*}, Kumar, R., Botting, J., Diana, V., Yu, H., Sen, A., Yan, J., Yildiz, F.[#], & Liu, J. (2026). Near-atomic in situ architecture and membrane-coupled dynamics of the *Vibrio cholerae* sheathed flagellum. **Nat. Commun.**, 17, 9199.
3. **Guo, W.**, Feng, L., Wang, Z., Guo, J., Park, D., Carroll, B. L., & Cheng, J. (2022). In situ high-resolution 3D imaging combined with proteomics and metabolomics reveals enlargement of subcellular architecture and enhancement of photosynthesis pathways in nuclear-irradiated *Chlorella pyrenoidosa*. **Chem. Eng. J.**, 430, 133037.
4. Yue, J.[#], **Guo, W.**[#], & Liu, J.[#] (2026). Integrating in situ single-particle cryo-electron microscopy with cryo-electron tomography for high-resolution structural biology in native cells. **Curr. Opin. Struct. Biol.**, 103373.
5. Zhang, K.^{*}, **Guo, W.**^{*}, Lynch, M. J., Crane, B. R., Liu, J., & Li, C. (2026). A MinD-like ATPase couples flagellation and cell division in spirochetes. *bioRxiv*, 2026.04.08.717139. **mBio**, accepted.
6. Yue, J.^{*}, Manne, A.^{*}, **Guo, W.**^{*}, Yu, H., Wang, J., Rahman, M. K., Lees, K., Xu, H., Botting, J. M., Motaleb, M. A., & Liu, J. (2026). In situ structure of the flagellar export apparatus in *Borrelia burgdorferi*. *bioRxiv*, 2026.05.13.724734. **Nat. Commun.**, accepted.
7. Wang, J.^{*}, Kurniyati, K.^{*}, **Guo, W.**^{*}, Botting, J. M., Sindelar, C. V., Li, C., & Liu, J. (2026). Asymmetric architecture and adaptation of *Treponema* flagella. **Nat. Commun.**, 17, 7834.
News: 'Atomic zoom' brings gum disease bacteria into sharp focus. Yale West Campus.
8. Yi, M., Guo, J., Zia, A., **Guo, W.**, Tachiyama, S., Ashton-Rickardt, G., Tan, W., Qiao, Y., Gong, Y., Egelman, E., Liu, J., Wang, F., & Xu, B. (2026). Cryo-structural insights into enzymatic peptide self-assembly driving extrinsic lytic cell death. **J. Am. Chem. Soc.**, 148(13), 14117–14128.
9. Kumar, R., Tachiyama, S., Yu, H., Heydari, S., Guo, J., Botting, J. M., **Guo, W.**, Hoover, T. R., & Liu, J. (2026). Assembly and glycosylation of *Helicobacter pylori* sheathed flagella. **PNAS Nexus**, 5(2), pgag011.
10. Bhattarai, N., **Guo, W.**, Partridge, J. D., & Harshey, R. M. (2025). Texas 2-step: a new model for YcgR::c-di-GMP action at the flagellar motor. **J. Bacteriol.**, e00353-25.
11. Sze, C. W., Zhang, K., Lynch, M. J., **Guo, W.**, Liu, J., Crane, B. R., & Li, C. (2025). The flagellin-specific chaperone FliS of *Borrelia burgdorferi* controls the cytoplasmic pool of flagellins at the level of translation initiation, secretion, and proteolysis. **Mol. Microbiol.**, 124(2), 173–187.
12. Yu, H., Wang, C., Yue, J., **Guo, W.**, Molineux, I. J., & Liu, J. (2025). Structural basis of bacteriophage λ infection initiation. **Sci. Adv.**, 11(46), eadw7914.
13. Botting, J. M., Rahman, M. K., Xu, H., Yue, J., **Guo, W.**, Del Mundo, J. T., Hammel, M., Motaleb, M. A., & Liu, J. (2025). FlbB forms a distinctive ring essential for periplasmic flagellar assembly and motility in *Borrelia burgdorferi*. **PLoS Pathog.**, 21(1), e1012812.
14. Yue, J., Heydari, S., Park, D., Chetrit, D., Tachiyama, S., **Guo, W.**, Botting, J. M., Wu, S., Roy, C. R., & Liu, J. (2025). In situ structures of the *Legionella* Dot/Icm T4SS identify the DotA-IcmX complex as the gatekeeper for effector translocation. **Proc. Natl. Acad. Sci. U.S.A.**, 122(39), e2516300122.
15. Zheng, W., Zhang, Y., Wang, J., Wang, S., ..., **Guo, W.**, Devarkar, S. C., & Wu, S. (2025). Visualizing the translation landscape in human cells at high resolution. **Nat. Commun.**, 16(1), 10757.
16. Kurniyati, K., Chang, Y., **Guo, W.**, Liu, J., Malkowski, M. G., & Li, C. (2023). Anti- σ^{28} factor FlgM regulates flagellin gene expression and flagellar polarity of *Treponema denticola*. **J. Bacteriol.**, 205(2), e00463-22.
17. Carroll, B. L., Nishikino, T., **Guo, W.**, Zhu, S., Kojima, S., Homma, M., & Liu, J. (2020). The flagellar motor of *Vibrio alginolyticus* undergoes major structural remodeling during rotational switching. **eLife**, 9, e61446.
18. **Guo, W.**^{*}, Guo, S.^{*}, Yue, J.^{*}, Chang, Y., Li, C., Motaleb, M. A.[#], & Liu, J.[#] (2026). In situ structural basis for how the bacterial flagellar motor switches rotational direction. **Manuscript in preparation.**
19. **Guo, W.**^{*}, Yue, J.^{*}, Li, C., Motaleb, M. A.[#], & Liu, J.[#] (2026). In situ near-atomic structure of the stator complex in the spirochete flagellar motor. **Manuscript in preparation.**

20. Yue, J. *, **Guo, W. ***, Motaleb, M. A., & Liu, J. (2026). In situ architecture of the *Borrelia burgdorferi* flagellar motor reveals structural adaptations for periplasmic motility. **Manuscript in preparation.**
21. Yue, J. *, **Guo, W. ***, Motaleb, M. A., & Liu, J. (2026). In situ structure of the *Borrelia burgdorferi* flagellar motor reveals FrbA-mediated reinforcement for high-torque rotation. **Manuscript in preparation.**
22. Park, J. H. *, **Guo, W. ***, Lee, C. K. *, Chen, J. W., Schmidt, W. C., Lele, P., Wong, G. C. L., Liu, J., & Yildiz, F. H. (2026). PlzC regulates flagellar motor function during surface attachment in a c-di-GMP-dependent manner. **Manuscript in preparation.**
23. Anderson, B. W., **Guo, W.**, Lim, B., Wang, J., Verdegaaal, A. A., Shen, C., Malvankar, N., Liu, J., & Goodman, A. L. (2026). Surface proteases shape the composition of the secretome in *Bacteroidales*. **Proc. Natl. Acad. Sci. U.S.A.**, submitted.

* Equal contribution; # corresponding/co-corresponding author.

Honors and Awards

2026	Best Poster Award , 11th Annual Yale Biophysics and Structural Biology Symposium
2025	Best Poster Award , Microbial Sciences Institute Retreat
2025	Best Poster Award , 11th New England Cryo-EM Symposium
2024	Gordon Research Conference on Sensory Transduction in Microorganisms Travel Grant
2022	Outstanding Youth of Sustainable Development, All America Youth Federation and Los Angeles
2022	Gordon Research Conference on Sensory Transduction in Microorganisms Travel Grant
2021	Zhejiang Province Award for Excellent Doctoral Graduate, China
2020	Zhejiang University Academic Award for Outstanding Doctoral Candidates, China
2019	Zhejiang University Outstanding Graduate Student, China

Talks and Posters

05/2026	Selected Speaker , 6th NY Area Cryo-EM Meeting, NYU Langone Medical Center. "Molecular mechanisms of sheathed flagellar assembly, rotation, and disassembly in <i>Vibrio cholerae</i> ."
04/2026	Speaker , YSM Structural Biology Seminar Series, Yale. "Multiscale in situ visualization of flagellar dynamics in <i>Vibrio cholerae</i> ."
03/2026	Job Talk , SMART, China. "Mapping bacterial nanomachine dynamics in situ at the molecular level."
01/2026	Discussion Leader and Poster Presenter , Gordon Research Conference, Sensory Transduction in Microorganisms, CA. "Mechanisms of the <i>Vibrio cholerae</i> sheathed flagellum."
10/2025	Selected Speaker , 12th New England Cryo-EM Symposium, UMass Chan Medical School. "How does the sheathed flagellar motor of <i>Vibrio cholerae</i> assemble and rotate?"
09/2025	Speaker , Annual Microbiology Retreat, Yale School of Medicine. "In situ cryo-EM structures reveal mechanisms of sheathed flagellar assembly, rotation, and disassembly in <i>Vibrio cholerae</i> ."
02/2025	Poster Presenter , Biophysical Society Annual Meeting, Los Angeles, CA. "Structural basis of the sheathed flagellar filament in <i>Vibrio cholerae</i> revealed by in situ cryo-EM single-particle analysis."
05/2024	Host , Yale Microbial Sciences Institute Small Talk, Yale. Invited Rasika Harshey, University of Texas at Austin, "C-di-GMP at the flagellar motor in <i>E. coli</i> / <i>Salmonella</i> ."
01/2024	Discussion Leader and Selected Speaker , Gordon Research Conference on Sensory Transduction in Microorganisms, California. "Capturing the rotational flagellar motor by cryo-electron tomography."
10/2023	Selected Speaker , 10th Semi-Annual New England Cryo-EM Symposium, Yale. "Capturing and modeling the structural dynamics of the bacterial flagellar motor."
10/2023	Host , Yale Microbiology Seminar Series. Invited Benjamin Engel, University of Basel, "Exploring the molecular architecture of photosynthesis and carbon fixation with cryo-ET."

- 03/2023 **Speaker**, Yale Microbial Sciences Institute Small Talk, Yale. “In situ structural basis of flagellar swimming and surface motility by pathogenic bacteria.”
- 11/2022 **Selected Speaker**, 9th Semi-Annual New England Cryo-EM Symposium, Harvard Medical School, Boston. “Capturing the structural dynamics of the bacterial flagellar motor.”
- 10/2022 **Speaker**, Postdoctoral Research in Progress Seminar Series, Department of Microbial Pathogenesis, Yale School of Medicine. “In situ structural basis of flagellar-based motility.”
- 09/2022 **Selected Speaker**, Gordon Research Conference on Sensory Transduction in Microorganisms, California. “In situ structural basis of how bacteria control flagellar motility.”
- 11/2021 **Selected Speaker**, 7th Semi-Annual New England Cryo-EM Symposium (virtual). “Combination of FIB-SEM and cryo-FIB achieves in situ 3D imaging of the subcellular architecture in *Chlorella pyrenoidosa*.”

Ad Hoc Journal Reviewer and Editor

- 2019–2026 Reviewer: *Chemical Engineering Journal*, *Bioresource Technology*, *Water Management*, and *Algal Research*
- 05/2026 Guest Associate Editor, *Frontiers in Microbiology*
- 02/2022 Guest Associate Editor, *Frontiers in Marine Science*
- 10/2022 Guest Associate Editor, *Frontiers in Bioengineering and Biotechnology*

Membership and Service

- 2025 Member, American Society for Microbiology
- 2023 Committee Member, Yale Annual Microbiology Retreat
- 2023 Committee Member, Yale Microbial Sciences Institute Annual Retreat
- 2022 Committee Member, Yale Microbiology Seminar Series
- 2022 Committee Member, Yale Microbial Sciences Institute Annual Postdoctoral Retreat

Complete List of Published Work

NCBI My Bibliography: <https://www.ncbi.nlm.nih.gov/myncbi/wangbiao.guo.1/bibliography/public/>